

Slides 5.3: Fibers, Natural and Manufactured

Slide outline for Lesson 5.3, one day, thirteen slides. Slides 10 and 11 are the burn demonstration and the no-flame version; use whichever one the building allows. *[Sal: get the open-flame answer in writing before you plan on slide 10.]*

Slide 1: What is this made of?

- Lesson 5.3
- Today you put your hands on six fabrics and figure out what they are
- Three tests: feel, stretch, water
- One teacher demonstration with a small flame, if the building allows it Image: six labeled fabric swatches laid out in a row on a table, letters A to F on masking tape.

Slide 2: Do now

- cotton, wool, silk, linen, polyester, nylon, rayon, spandex
- Four come from a plant or an animal. Four are made in a factory.
- Sort them into two lists. Guess if you do not know.
- Two minutes Image: two empty columns headed “from a plant or animal” and “made in a factory.”

Slide 3: The answer, and the tricky one

- Natural: cotton, wool, silk, linen
- Manufactured: polyester, nylon, rayon, spandex
- Rayon starts as wood pulp, which is a plant, but it is made into fiber in a factory. It counts as manufactured. Image: a cotton boll, a sheep, a silkworm cocoon, and a flax plant on the left; a factory with spools coming out of it on the right.

Slide 4: One of these is a plastic bottle

- A fleece jacket and a wool sweater
- Both keep you warm
- One is made from plastic. One is made from a haircut.
- By the end of today you can tell them apart with your hands Image: a fleece jacket and a wool sweater side by side, with a plastic bottle next to the fleece.

Slide 5: Natural fibers, what each one is good and bad at

- Cotton: soft, absorbent, breathable. Wrinkles. Shrinks. Slow to dry.
- Wool: warm even when damp, springs back. Itchy to some people. Hot water felts it.
- Silk: strong and smooth, costly. Water can mark it. Hand wash.
- Linen: coolest in heat, gets softer with age. Wrinkles the most. Image: four swatches labeled cotton, wool, silk, linen with a small source icon on each.

Slide 6: Manufactured fibers, what each one is good and bad at

- Polyester: strong, holds shape, dries fast, cheap. Does not breathe. Holds odor. Melts.
- Nylon: very strong and slick. Windbreakers and tights. Melts.
- Rayon: drapes like silk, from wood pulp. Weak when wet.
- Spandex: stretches and comes back, always blended. Heat destroys it. Image: four swatches labeled polyester, nylon, rayon, spandex with a small factory icon on each.

Slide 7: The class table

Fiber	Warmth	Water	Wrinkles	Stretch	Cost and care
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- Fill this in on the handout as we go
- The fiber decides the care. The care decides how long the garment lasts. Image: the blank five-column table, large enough to read from the back. Notes: Say that last line twice. It is the bridge to tomorrow's care label lesson.

Slide 8: The three tests

1. Feel: rub it between finger and thumb. Smooth or rough? Warm or cool? Thick or thin?
2. Stretch: pull gently across and let go. Does it come back?
3. Water: one drop. Count to ten. Soaked in, beaded up, or spread out? Image: three small drawings, fingers rubbing cloth, two hands pulling cloth, and a dropper releasing one drop onto cloth.

Slide 9: Station rules

- One drop, not a puddle
- Swatches stay at the station
- Wet swatches go on the paper towel, letter side up
- Ninety seconds per swatch. Six swatches.
- Nothing goes in a mouth Image: a station setup photo: six swatches, a water cup, a dropper, a paper towel, and a sealed envelope.

Slide 10: The burn test, teacher demonstration only

- I hold the tweezers. The flame is on for under three seconds.
- Nobody stands up. Nobody films it.
- The extinguisher is for adults. PASS is pull, aim at the base, squeeze, sweep. Students get out and tell the teacher.
- Watch three things: what the flame does, the smell, what is left behind Image: a metal pie pan on a table with long tweezers holding a small fabric snip, and a taped line on the floor with chairs behind it. Notes: Same fire rules, same words, as Lesson 1.3. Tell the class the day before that there will be a flame, so anyone who needs to sit out can tell you quietly.

Slide 11: Three fibers, three results

- Cotton: burns fast, smells like burning paper, soft gray ash that crumbles

- Wool: burns slowly, curls away, smells like burning hair, a black bead you can crush, often puts itself out
- Polyester: does not burn, it melts and shrinks away, smells sweet and chemical, leaves a hard bead
- A fiber that melts sticks to skin. That is why firefighters and welders do not wear fleece. Image: three small results side by side, gray crumbly ash, a black crushable bead, and a hard shiny bead. Notes: If the building says no flame: play the same three burns on video, then run the wrinkle test (fist for ten seconds, rate 1 to 3) and make the melt-versus-burn point from the video.

Slide 12: Open the envelope

- Which letter was which fiber?
- Count how many you got right
- The one nobody gets is usually silk against polyester Image: a sealed envelope being opened with six letters and fiber names listed inside.

Slide 13: Pick the fabric for the job

- A winter hat
- A soccer jersey
- A dress shirt for a job interview
- A swimsuit
- Two pieces of evidence from your own tests. Use the words: absorbed, beaded, came back, stayed stretched. Image: four small icons, a knit hat, a jersey, a collared shirt, and a swimsuit. Notes: 3-2-1 exit card. Tomorrow: the tag inside the garment, which is where the fiber and the care rules are printed.